PROBLEM STATEMENT · DATA TO ACTION

Supply Chain Risk Detection & Decision-Support System

From raw operational data to prioritised risks, assigned owners and automated action, built for Savanna Distribution Ltd.

01 · BUSINESS CONTEXT

The problem is not a lack of data.

Savanna Distribution Ltd is a fictional East African distributor managing products across multiple warehouses, suppliers and customers. It processes orders, manages inventory, buys from suppliers and delivers to customers. As the business grows, so does its data. The challenge is turning that data into timely decisions.

02 · THE BUSINESS PROBLEM

Teams discover problems after the damage is done.

Silent stockouts

A product runs out before procurement realises inventory is critical.

Drifting suppliers

A supplier slows down, but it is noticed only after several late deliveries.

Late backorders

A customer order becomes backordered and the team finds out too late.

Imbalanced stock

One warehouse holds excess inventory while others face shortages.

No clear priority

Many risks exist at once, with no way to decide which comes first.

No clear owner

Even when a risk is found, who should act is unclear.

The result is a gap between **data and action**. The business needs a system that identifies risks early, quantifies impact, prioritises, assigns responsibility and triggers a response.

03 · THE CORE BUSINESS QUESTION

One question drives the system.

> Which supply-chain risks require attention right now, how serious are they, what could they cost the business, and who needs to act?

04 · KEY BUSINESS QUESTIONS

What the system must answer.

Inventory

Which products may stock out? How many days of cover remain, and will it last until replenishment? Which stockouts cost most? Where is stock excessive?

Suppliers

Who is becoming less reliable versus their baseline? Who is chronically late? Which open purchase orders are exposed to delay?

Customer Orders

Which orders are late or backordered, carry significant financial impact, or need immediate intervention?

Decision-making

What is handled first? What is the financial exposure? Who owns each risk, and what action should they take?

05 · EVIDENCE FROM THE RAW DATA

It starts with understanding the data, not a dashboard.

168,224
deduplicated order lines

KES 967M
total order value

6
realistic data-quality issues

- Duplicate records
- Inconsistent order statuses
- Incorrectly formatted dates
- Delivered orders without delivery dates
- Invalid purchase order date relationships
- Negative inventory values

These issues are intentional. Real business data is rarely clean when it first arrives.

06 · SUCCESS CRITERIA

The system should be able to:

1. Identify meaningful risks from historical and current operational data.
2. Estimate the potential financial impact of each risk.
3. Assign an urgency score.
4. Prioritise risks into P1, P2 and P3 levels.
5. Assign each risk to the appropriate business owner.
6. Recommend an action.
7. Present risks in a management-facing Power BI dashboard.
8. Notify responsible owners through the automation layer.

The goal is not the maximum number of alerts. It is **useful alerts that help people decide what to do next.**

07 · BUSINESS OUTCOME

From reacting to anticipating.

Reactive problem solving

Discover the stockout after it happens.

→

Proactive risk management

Spot the risk early enough to respond.

Instead of showing that a supplier is late, the system answers: is this supplier getting worse, how significant is the exposure, and who should intervene? Instead of hundreds of equal alerts, it surfaces the few that matter most.

08 · FROM DATA TO ACTION

Eight layers, one decision flow.

01
Data
Tells us what happened.

02
Evidence
Turns raw data into business signals.

03
Risk
Identifies potential problems.

04
Impact
Estimates what it could cost.

05
Urgency
How quickly to respond.

06
Priority
Which to handle first.

07
Owner
Who is responsible.

08
Action
A practical next step.

09 · SOLUTION OVERVIEW

Architecture.

Raw Data → Silver → Gold → Intermediate → Risk Engine → Risk Alerts → Power BI → n8n → Email → Human Action

Databricks

Processes operational data, builds business-ready datasets, calculates risk indicators and generates prioritised alerts.

Power BI

The Supply Chain Risk Command Centre for monitoring the current risk landscape.

n8n

The automation layer that routes risk information to the right business owner.

Email

The operational notification, so important risks never stay hidden inside a dashboard.

10 · WHAT THIS PROJECT DEMONSTRATES

The workflow matters more than the tools.

Raw operational data | Business-ready data | Risk detection | Decision support | Operational automation

A BI system that goes beyond reporting history, to identify, prioritise and respond to operational risk.

PROJECT PHILOSOPHY

Don't build a dashboard just to show data. Build a system that helps someone make a decision.

Data → Analysis → Risk → Decision → Action